

1 The multiplicative chain rule

Source: PrimeTensor/Analysis/Chain.lean

What this file establishes

The chain rule, under hypotheses the module header is unusually explicit about. Differentiability of the two factors and scale control of their tangent maps are, in its words, *almost* sufficient; one further hypothesis is exposed rather than hidden, namely that negligibility measured against the actual inner response germ implies negligibility measured against the original perturbation. That is ChainAdmissibleAt, and Design note 1.9 is about why it has to be assumed here when its classical counterpart is automatic.

Around it the file assembles: composition of tangent maps; the fact that a differentiable map with a controlled tangent has a response that really does approach the pivot, so that the inner response is itself a legitimate perturbation; and an exact identity saying the response of a composite is the outer response along the inner response. The proof of the rule is then a telescoping product of two ratios, and Design note 1.13 sets it beside the classical two-term error split.

1.1 Ratio algebra

Lemma 1.1 (MulReal.ratio_comp, ratio_mul_right and mul_ratio_right). $\text{ratio}(a, c) = \text{ratio}(a, b) \text{ratio}(b, c)$, and $\text{ratio}(a, b) \cdot b = a = b \cdot \text{ratio}(a, b)$.

```
theorem ratio_mul_right (a b : MulReal) :
  ratio a b * b = a := by
  unfold ratio
  calc
    (a * b⁻¹) * b = a * (b⁻¹ * b) := mul_assoc a b⁻¹ b
    _ = a * 1 := by rw [inv_mul b]
    _ = a := mul_one a
```

Proof. The first is the argument of PrimeTensor/Scale/Laws.lean one layer up: reassociate to expose $b^{-1}b$, cancel, drop the unit. The second is the same cancellation with the intermediate magnitude restored, and the third is the second after commuting. $\square$

The cocycle identity and the reconstruction identity are the two facts the main proof runs on. The first telescopes a chain of comparisons; the second says that a magnitude is recoverable from a comparison and its reference point, which is what lets the inner response be treated as a perturbation in its own right.

Lemma 1.2 (MulReal.converges_congr and negligible_congr). *Convergence and negligibility are unchanged by replacing a germ with one equal to it term by term.*

Proof. Keep the anchor and rewrite pointwise. $\square$

These are bookkeeping, but they carry real weight below: several of the germs in the main proof are equal on the nose rather than merely asymptotic, and these two lemmas are how an exact rewriting is admitted into a statement about levels.

1.2 Composition of tangent maps

Definition 1.3 (`MulTangentMap.comp`). $\text{comp}(D_g, D_f)(h) \equiv D_g(D_f(h))$, with the three morphism laws proved by rewriting with the corresponding laws of the two factors, and `comp_apply` recording the value by `rfl`.

```
def comp (Dg Df : MulTangentMap) : MulTangentMap where
  toFun := fun h => Dg (Df h)
  map_one := by
    rw [Df.map_one, Dg.map_one]
  map_mul := by
    intro a b
    rw [Df.map_mul, Dg.map_mul]
  map_inv := by
    intro a
    rw [Df.map_inv, Dg.map_inv]
```

Lemma 1.4 (`MulTangentMap.map_ratio`). *Every tangent map preserves ratios: $D(\text{ratio}(a, b)) = \text{ratio}(D(a), D(b))$.*

Proof. Unfold the ratio and apply preservation of products and of inverses. $\square$

This is what lets an error, which is a ratio, be pushed through a tangent map and still be recognised as an error between the images.

1.3 Differentiability gives a genuine perturbation

Lemma 1.5 (`negligible_converges`). *If error is negligible relative to base and base approaches the pivot, then error converges to the pivot.*

Proof. Fix a level. Take the anchor negligibility supplies at the smallest gain `one`, and the anchor the convergence of the base supplies at the requested level, and join them. Beyond the join the base achieves that level, so negligibility puts the error at `advance(ℓ, one)`, which is `succ ℓ`; one weakening by `PrimeTensor/Analysis/Control.lean` gives `ℓ`. $\square$

Theorem 1.6 (`response_converges`). *If f has derivative D at x , D is scale controlled, and h approaches the pivot, then $\text{response}(f, x, h)$ converges to the pivot.*

Proof. The response is recovered from its error and the model by Lemma 1.1:

$$\text{response}_n = \text{ratio}(\text{response}_n, D(h_n)) \cdot D(h_n).$$

The first factor converges to the pivot by Lemma 1.5, since differentiability makes it negligible relative to h and h approaches the pivot. The second converges to the pivot by `maps_converges` of `PrimeTensor/Analysis/Control.lean`, which is where control of D is used. Their product converges to $1 \cdot 1 = 1$ by `PrimeTensor/Analysis/Near/Laws.lean`, and Lemma 1.2 transports that along the pointwise identity above. $\square$

Remark 1.7 (Differentiable implies continuous — but not for free). This is the statement that a differentiable map is continuous at the point, in the only form the vocabulary offers: the response along any perturbation is itself a perturbation. In the additive theory it needs no extra hypothesis. Here it needs D to be scale controlled, and unavoidably so —

`PrimeTensor/Analysis/Derivative.lean` placed no regularity on tangent maps, so an unbounded model could carry a perturbation away from the pivot and the response with it. Control is what excludes that, and this theorem is the first place the notion is spent rather than established.

1.4 The admissibility hypothesis

Definition 1.8 (`ChainAdmissibleAt`). f is *chain admissible at x* when, for every h approaching the pivot, every germ negligible relative to $\text{response}(f, x, h)$ is negligible relative to h .

```
def ChainAdmissibleAt
  (f : MulReal → MulReal) (x : MulReal) : Prop :=
  ∀ h : MulReal.Seq,
    ApproachesPivot h →
    ∀ error : MulReal.Seq,
      NegligibleRelative error (response f x h) →
      NegligibleRelative error h
```

Design note 1.9 (Why this is a hypothesis and not a lemma). The header identifies the classical counterpart: the comparison $Df(h) = O(h)$ that transports an outer little- o error back to the original variable. Classically that comparison is not assumed — for a differentiable map with a bounded derivative it follows, since the increment is bounded by a constant times the perturbation.

It is not lifted here in that form. The obstacle is one already recorded in `PrimeTensor/Analysis/Derivative.lean`: comparisons in that file are made with *advance*, which refines a level by a *positive* number of steps, there being no advance by zero. Used that way the vocabulary can say “strictly finer by a prescribed amount” — little- o — and cannot say “no coarser than, up to a constant” — big- O ; and the statement the chain rule needs is a big- O statement. Nor could the missing inequality simply be written out, since `Depth` has no subtraction with which to say that one level lags another by a bounded amount.

`ChainAdmissibleAt` is what this file writes instead. Rather than bounding the inner response against the perturbation, it asserts the consequence that bounding would have: that the two germs are interchangeable as *bases* for negligibility, in the direction the proof consumes. This converts an inequality the file does not express into an implication it does, at the cost of making it an assumption. Whether it follows from the other hypotheses of Theorem 1.12 is not settled here, and no argument either way is given.

The obstacle is, however, one of formulation rather than of expressive power. `PrimeTensor/Analysis/Growth.lean` writes the growth comparison after all, by advancing *both* levels — the base one and the response one — and comparing the results in the tail order instead of demanding a fixed refinement of a single level. That sidesteps the missing subtraction, and the condition so obtained implies the hypothesis assumed here.

Lemma 1.10 (`chainAdmissibleAt_identity`). *The identity map is chain admissible at every point.*

Proof. Its response along h at stage n is $\text{ratio}(xh_n, x)$. That is h_n , by the reconstruction identity of `PrimeTensor/Analysis/Derivative.lean`. So the two bases are the same germ and the implication is trivial: rewrite the base and hand back the hypothesis. $\square$

1.5 Responses of a composite

Lemma 1.11 (`response_comp`). *For every n ,*

$$\text{response}(g \circ f, x, h)_n = \text{response}(g, f(x), \text{response}(f, x, h))_n.$$

```
theorem response_comp
  (f g : MulReal → MulReal)
  (x : MulReal)
  (h : MulReal.Seq) :
  ∀ n : Depth,
    response (fun y => g (f y)) x h n =
      response g (f x) (response f x h) n := by
intro n
unfold response
rw [MulReal.mul_ratio_right
  (f (x * h n))
  (f x)]
```

Proof. The left side is $\text{ratio}(g(f(xh_n)), g(f(x)))$. The right side is $\text{ratio}(g(f(x) \cdot \text{inner}_n), g(f(x)))$ with $\text{inner}_n = \text{ratio}(f(xh_n), f(x))$, and Lemma 1.1 reconstructs $f(x) \cdot \text{inner}_n = f(xh_n)$. The two sides are then the same term. $\square$

An identity, with no error and no hypothesis on h : the perturbation of the outer function is exactly the response of the inner one. This is what makes the chain rule a matter of composing two first-order statements rather than of estimating a composite directly.

1.6 The chain rule

Theorem 1.12 (`hasMulDerivativeAt_comp`). *Suppose f has derivative D_f at x with D_f scale controlled, f is chain admissible at x , and g has derivative D_g at $f(x)$ with D_g scale controlled. Then $y \mapsto g(f(y))$ has derivative $\text{comp}(D_g, D_f)$ at x .*

```
theorem hasMulDerivativeAt_comp
  {f g : MulReal → MulReal}
  {x : MulReal}
  {Df Dg : MulTangentMap}
  (hf : HasMulDerivativeAt f x Df)
  (hDf : Df.ScaleControlled)
  (hadm : ChainAdmissibleAt f x)
  (hg : HasMulDerivativeAt g (f x) Dg)
  (hDg : Dg.ScaleControlled) :
  HasMulDerivativeAt
    (fun y => g (f y))
    x
    (MulTangentMap.comp Dg Df) := by
```

Proof. Fix h approaching the pivot and write $\text{inner} = \text{response}(f, x, h)$.

By Theorem 1.6, inner approaches the pivot, so it is a legitimate perturbation and the hypothesis on g may be applied *along it*. That gives the outer error $\text{ratio}(\text{response}(g, f(x), \text{inner})_n, D_g(\text{inner}_n))$ negligible relative to inner ; chain admissibility re-bases it against h .

Separately, the hypothesis on f gives the inner error $\text{ratio}(\text{inner}_n, D_f(h_n))$ negligible relative to h . Push it through D_g using `negligible_map` of `PrimeTensor/Analysis/Control.lean` — this is where control of D_g is used — and rewrite the image with Lemma 1.4, obtaining $\text{ratio}(D_g(\text{inner}_n), D_g(D_f(h_n)))$ negligible relative to h .

Multiply the two negligible germs, which is legitimate by `PrimeTensor/Analysis/Rules.lean`. Their product telescopes: by the cocycle identity of Lemma 1.1,

$$\text{ratio}(\text{resp}_n, D_g(\text{inner}_n)) \cdot \text{ratio}(D_g(\text{inner}_n), D_g(D_f(h_n))) = \text{ratio}(\text{resp}_n, D_g(D_f(h_n))).$$

Finally Lemma 1.11 identifies resp_n with the response of the composite, and $\text{comp}(D_g, D_f)(h_n) = D_g(D_f(h_n))$ by `rfl`, so the product is exactly the error of the composite against its model. $\square$

Design note 1.13 (A telescope where the classical proof has a sum). The classical chain rule splits the error into two pieces,

$$\begin{aligned} g(f(x+h)) - g(f(x)) - D_g D_f h &= [g(f(x+h)) - g(f(x)) - D_g \Delta] + D_g [\Delta - D_f h], \\ \Delta &= f(x+h) - f(x), \end{aligned}$$

and estimates the two summands separately, the second using linearity and boundedness of D_g . The proof above has the same two pieces and combines them differently: they are ratios, they are *multiplied*, and the combination is the cocycle identity, which is exact. There is no triangle inequality and no constant — the intermediate term $D_g(\text{inner}_n)$ cancels identically rather than being bounded away.

Each of the three hypotheses is spent in exactly one place, and the proof makes that visible: control of D_f only to know the inner response is a genuine perturbation; control of D_g only to push the inner error through it; admissibility only to re-base the outer error. The docstring says as much.

Remark 1.14 (The rule cannot yet be iterated). Applying Theorem 1.12 twice requires knowing that $\text{comp}(D_g, D_f)$ is itself scale controlled, and that is proved nowhere: `PrimeTensor/Analysis/Control.lean` closed the property under product and inverse, but composition was not defined until this file, and no lemma here adds it. Likewise `ChainAdmissibleAt` is not shown closed under composition. So the rule is stated for one composite and has no corollary for a tower of them.

Remark 1.15 (Only one function is known to be admissible). Lemma 1.10 is the sole verification of `ChainAdmissibleAt` in the file. Nothing establishes it for a constant map, for a power, or for a product — so the only f with which Theorem 1.12 can currently be discharged is the identity, where it says nothing. This continues the pattern noted since `PrimeTensor/Analysis/Derivative.lean`: the machinery is complete and its nontrivial instances are still ahead.

Remark 1.16 (What is not proved). Admissibility is not shown to follow from differentiability and control, nor shown independent of them. There is no uniqueness of derivatives, so “the” derivative of a composite is still not a phrase the development can use. `comp` is given no laws — not associativity, and not that identity is a unit for it — and no relation is stated between `comp` and the pointwise product of `PrimeTensor/Analysis/Rules.lean`.

Summary of the correspondence

Lean declaration	Kind	Here
<code>MulReal.ratio_comp</code>	theorem	Lem. 1.1
<code>MulReal.ratio_mul_right</code>	theorem	Lem. 1.1
<code>MulReal.mul_ratio_right</code>	theorem	Lem. 1.1
<code>MulReal.converges_congr</code>	theorem	Lem. 1.2
<code>MulTangentMap.comp</code>	definition	Def. 1.3
<code>MulTangentMap.comp_apply</code>	simp thm	Def. 1.3
<code>MulTangentMap.map_ratio</code>	theorem	Lem. 1.4
<code>MulDifferential.negligible_congr</code>	theorem	Lem. 1.2
<code>negligible_converges</code>	theorem	Lem. 1.5
<code>response_converges</code>	theorem	Thm. 1.6
<code>ChainAdmissibleAt</code>	definition	Def. 1.8
<code>chainAdmissibleAt_identity</code>	theorem	Lem. 1.10
<code>response_comp</code>	theorem	Lem. 1.11
<code>hasMulDerivativeAt_comp</code>	theorem	Thm. 1.12

Every declaration of `PrimeTensor/Analysis/Chain.lean` appears above. The file contains no sorry, no axiom, and no unproved named proposition. Design notes 1.9 and 1.13 and Remarks 1.7, 1.14, 1.15 and 1.16 are stated for the reader and are not declarations of the file; in particular the comparison with the classical error split is a gloss and is not formalised.